Python & Computer Networks Practice Worksheet

Section A – 2 Mark Questions

1. Function Return Statement

(a) What is the purpose of the `return` statement in a Python function? Illustrate with a small example.

(b) Explain what happens if a function does not contain a `return` statement. Give an example of a function that calculates the cube of a number and returns the result.

2. Type Conversion

(a) Write one example each of explicit type conversion and implicit type conversion in Python.

(b) Differentiate between explicit and implicit type conversion with one example of each.

3. Dictionary Operations

Practice 1: Consider the dictionaries:

```
D = {"Aman": 45, "Bina": 60, "Chirag": 75}
D1 = {"Deepa": 50, "Esha": 85, "Aman": 30}
```

(a) Write a statement to display the value corresponding to the key `"Bina"` in dictionary `D`. Also write a statement to display the length of dictionary `D1`.

(b) Write a statement to append all key-value pairs of dictionary `D` into dictionary `D1`. Also write a statement to delete the item with key `"Esha"` from dictionary `D1`.

Practice 2: Consider the dictionaries:

```
Dict_A = {"Apple": 100, "Mango": 150}
Dict_B = {"Banana": 60, "Apple": 80, "Cherry": 120}
```

(a) Write a statement to display the value corresponding to the key `"Mango"` in `Dict_A`. Also write a statement to display the length of `Dict_B`.

(b) Write a statement to append all key-value pairs of `Dict_A` into `Dict_B`. Also write a statement to delete the item with key `"Banana"` from `Dict_B`.

4. Random Module Output

Practice 1: What possible output is expected from the given code?

```
import random
Colors = ["Red", "Blue", "Green", "Yellow"]
for i in range(2):
    print(Colors[random.randint(1, i+2)], end="#")
```

(A) Blue#Yellow# (B) Blue#Red# (C) Yellow#Green# (D) Red#Blue#

Practice 2: What possible output is expected from the given code?

```
import random
Fruits = ["Apple", "Mango", "Banana", "Orange"]
for i in range(2):
    print(Fruits[random.randint(0, i+1)], end="&")
```

(A) Mango&Banana& (B) Apple&Orange& (C) Banana&Apple& (D) Orange&Mango&

5. Error Correction

Practice 1: The following code accepts `N` as an integer argument and returns the sum of all integers from 1 to N. Rewrite it after removing all syntax and logical errors. Underline all corrections made.

```
def Total (N)
for I in range (N) :
S=S+I
return S
print (Total (8)
```

Correction Expected: **def Total(N):**
S = 0
for I in range(1, N+1):
S = S + I
return S
print(Total(**8**))

Practice 2: The following code is intended to accept a positive integer `N` and return the sum of its digits (e.g., for 123, it should return 6). Rewrite it after removing all syntax and logical errors. Underline all corrections made.

```
def SumDigits (N):
Sum = 0
while N > 0
    d = N % 10
    Sum = Sum + d
    N = N / 10
return Sum
print (SumDigits("456")
```

Correction Expected:

```
def SumDigits(N):
    Sum = 0
    while N > 0:
        d = N % 10
        Sum = Sum + d
        N = N // 10
    return Sum
print(SumDigits(456))
```

6. Networking Terms

Practice 1:

(a) Expand and explain the term URL.

(b) Expand the term PPP. What is its use?

Practice 2:

(a) Expand and explain the term HTML.

(b) Expand the term VoIP. What is its primary use?

7. Network Topology & Protocols

Practice 1:

(a) List one advantage and one disadvantage of Star topology.

(b) What is a protocol in the context of computer networks? Which protocol is used to transmit hypertext across the web?

Practice 2:

(a) List one advantage and one disadvantage of Bus topology.

(b) What is a protocol in the context of computer networks? Which protocol is used for transferring files over the internet?

Section B – 3 Mark Questions

8. File Handling – Line Search

Practice 1: Write a Python function that displays all the lines containing the word `'price'` from a text file `"Items.txt"`.

For example, if the file contains:

```
The price of the item is fixed.
Discounts are offered on selected items.
Every price tag must be verified.
```

Then the output should be:

```
The price of the item is fixed.
Every price tag must be verified.
```

Practice 2: Write a Python function that displays all the lines containing the word `'vote'` from a text file `"Elections.txt"`.

For example, if the file contains:

```
In an election many people vote to choose their representative.
The candidate getting the maximum share of votes stands elected.
Normally, one person has to vote once.
```

Then the output should be:

```
In an election many people vote to choose their representative.
Normally, one person has to vote once.
```

9. File Handling – Digit Count / Word Search

Practice 1: (a) Write a Python function that counts and returns the number of digits appearing in the text file `"Data.txt"`.

For example, if the file contains:

```
Python 3.12 was released in 2023 with 5 new features.
Version 2.7 is still used by 12 percent of users.
```

Then the function should return `11` (count of digit characters).

Practice 2: (b) Write a Python function that displays the words in which the lowercase letter `'a'` appears at least twice in the text file `"Data.txt"`.

For example, if the file contains:

```
Banana and papaya are available at the market.
A cat sat on a mat near the canal.
```

Then the function should display:

```
Banana papaya available
```

10. Code Output Tracing

Practice 1: Write the output of the following code:

```
def Practice2027(given):
    new = []
    for ch in given[1:-1]:
        if ch.islower():
            new.reverse()
        elif ch not in new:
            new.append(ch)
        elif ch in new:
            new.pop()
    print(new)

Practice2027("Silver-36Points")
```

Practice 2: Write the output of the following code:

```
def MysteryFunc(given):
    result = []
    for ch in given[1:-1]:
        if ch.isupper():
            result.reverse()
        elif ch not in result:
            result.append(ch)
        elif ch in result:
            result.pop()
    print(result)

MysteryFunc("Gold-24Medals")
```

Section C – 4 Mark Questions

11. CSV File Handling – Filter and Write

Practice 1: A CSV file `"Cities.csv"` contains data about cities. Each record contains:
`City Name, State, Population, Language`

For example: `['Jaipur', 'Rajasthan', 3046163, 'Hindi']`

Write a Python program that reads data from this file and appends all records where population is more than `5000000` into another CSV file `"BigCities.csv"`.

Note: `"Cities.csv"` also contains a header row. The header row should NOT be copied to `"BigCities.csv"`.

Practice 2: A CSV file `"States.csv"` contains data about states. Each record contains:
`State Name, Capital, Population, Language`

For example: `['Karnataka', 'Bengaluru', 61095297, 'Kannada']`

Write a Python program that reads data from this file and appends all records where population is less than `10000000` into another CSV file `"SmallStates.csv"`.

Note: `"States.csv"` also contains a header row. The header row should NOT be copied to `"SmallStates.csv"`.

12. CSV File Handling – User-Defined Functions

Practice 1: Meera runs a stationery shop. She maintains a CSV file `"Orders.csv"` with columns:
`Item_ID, Item_Name, Quantity, Price_Per_Item`

Write user-defined functions:

I. `AddItem()` – to accept an order record from the user and add it to `Orders.csv`.

II. `TotalBill()` – to calculate and return the total bill based on `Quantity` and `Price_Per_Item`.

Practice 2: Raj is the manager of a medical store. He maintains a CSV file `"Sales.csv"` with columns:
`Product_ID, Product_Name, Quantity_Sold, Price_Per_Unit`

Write user-defined functions:

I. `Accept()` – to accept a sales record from the user and add it to `Sales.csv`.

II. `CalculateTotalSales()` – to calculate and return the total sales based on `Quantity_Sold` and `Price_Per_Unit`.

13. Binary File Handling – Create and Search

Practice 1: `"STUDENTS.DAT"` is a binary file storing records with structure:
`[RollNo, Name, City, Marks]`
where RollNo is string, Name and City are strings, and Marks is float.

Write user-defined functions:

(i) `Create()` – to input data for students and write it to the binary file `STUDENTS.DAT`.

(ii) `SearchCity(C)` – to read contents from the file and display details of students whose `City` matches with the value of `C`.

Practice 2: `"PASSENGERS.DAT"` is a binary file storing records with structure:
`[PNR, PName, BRDSTN, DESTN, FARE]`
where PNR, PName, BRDSTN, DESTN are strings, and FARE is float.

Write user-defined functions:

(i) `Create()` – to input data for passengers and write it to the binary file `PASSENGERS.DAT`.

(ii) `SearchDestn(D)` – to read contents from the file and display details of passengers whose `DESTN` matches with the value of `D`.

14. Binary File Handling – Tuple Records

Practice 1: An institute stores data of its trainers in a binary file `"TRAINERS.DAT"` using the following record structure (each record is a tuple):
`(T_ID, T_Name, T_Subject, Fee)`

where:

- `T_ID` – Trainer ID (integer)
- `T_Name` – Trainer Name (string)
- `T_Subject` – Subject of expertise (string)
- `Fee` – Fee per session (float)

Write user-defined functions:

(i) `AddTrainer()` – to input data for trainers and write it to the binary file `TRAINERS.DAT`.

(ii) `ShowTrainers(S)` – to read contents from the file and display details of trainers whose `T_Subject` matches with the value of `S`.

Practice 2: NextStep is an organization that stores data of its Resource Persons in a binary file `"RESOURCES.DAT"` using the following record structure (each record is a tuple):
`(R_ID, R_Name, R_Expertise, Charges)`

where:

- `R_ID` – Resource Person's ID (integer)
- `R_Name` – Resource Person's Name (string)
- `R_Expertise` – Area of expertise (string)
- `Charges` – Charges per hour (float)

Write user-defined functions:

(i) `AddResource()` – to input data for resource persons and write it to the binary file `RESOURCES.DAT`.

(ii) `SearchExpertise(E)` – to read contents from the file and display details of resource persons whose `R_Expertise` matches with the value of `E`.

Section E – 5 Mark Questions (Computer Networks Case Studies)

Practice Question 1: "HealthTech Solutions"

"HealthTech Solutions" is a healthcare IT company with its head office in Pune and a new branch in Nagpur. The Nagpur branch is setting up a new facility with four main blocks: **HR, FINANCE, IT, and SERVER**. As a network expert, suggest the best network solutions to resolve the issues mentioned in points (i) to (v), keeping the following parameters in mind:

From	To	Distance
HR	FINANCE	55 m
HR	IT	95 m
HR	SERVER	120 m
FINANCE	IT	45 m
FINANCE	SERVER	110 m
IT	SERVER	70 m

Distance of Nagpur Branch from Pune Head Office = 850 km

Block	No. of Computers
HR	25
FINANCE	60
IT	110
SERVER	15

Questions:

(i) Suggest the most appropriate location of the server inside the Nagpur Branch. Justify your choice.

(ii) Which hardware device will you suggest to connect all the computers within each block of the Nagpur Branch?

(iii) Draw a cable layout to efficiently connect various blocks within the Nagpur Branch.

(iv) Where should the router be placed to provide internet to all the computers in the Nagpur Branch?

(v) The manager wants to connect the Nagpur branch to the Pune Head Office for data sharing. Which transmission medium (wired/wireless) will be most suitable for this? Justify.

> **Help Notes for Practice:**
>
> - **(i) Server Location:** Place the server in the block with the *maximum number of computers* (to minimize cable length and maximize efficiency). For this case, **IT block (110 computers)** is the best location.
> - **(ii) Hardware Device within Blocks:** Use a **Switch** or **Hub** to connect computers within a single block. (Switch is preferred for better performance).
> - **(iii) Cable Layout:** Draw a graph connecting all blocks with the minimum total cable length. Start with the central block (IT) and connect it to the nearest blocks, then connect remaining blocks to the nearest already-connected block (Prim's/Kruskal's algorithm concept).
> - **(iv) Router Placement:** Place the router in the block where the server is located (IT block) or at the main entry point of the building, as it needs to connect the entire branch to the outside world.
> - **(v) Transmission Medium:** Since the distance is large (850 km), use a **wired medium (Optical Fiber)** for high speed and reliability, or **wireless (Satellite/Radio waves)** if laying cables is not feasible. Justify based on speed, cost, and distance.

Practice Question 2: "EduGlobal Academy"

"EduGlobal Academy" is a large educational institution with its main campus in Delhi and a new campus in Mumbai. The Mumbai campus is expanding and has four blocks: **ADMIN, LIBRARY, SCIENCE, and ARTS**. As a network expert, suggest the best network solutions to resolve the issues mentioned in points (i) to (v), keeping the following parameters in mind:

From	To	Distance
ADMIN	LIBRARY	40 m
ADMIN	SCIENCE	90 m
ADMIN	ARTS	130 m
LIBRARY	SCIENCE	50 m
LIBRARY	ARTS	100 m
SCIENCE	ARTS	60 m

Distance of Mumbai Campus from Delhi Campus = 1400 km

Block	No. of Computers
ADMIN	35
LIBRARY	20
SCIENCE	75
ARTS	45

Questions:

(i) Suggest the most appropriate location of the server inside the Mumbai Campus. Justify your choice.

(ii) Which hardware device will you suggest to connect all the computers within each block of the Mumbai Campus?

(iii) Draw a cable layout to efficiently connect various blocks within the Mumbai Campus.

(iv) Where should the router be placed to provide internet to all the computers in the Mumbai Campus?

(v) The academy wants to conduct live online lectures for students in both Delhi and Mumbai simultaneously. Which protocol will be used for this real-time audio/video communication over the internet?

Help Notes for Practice:

- **(i) Server Location:** Place the server in the block with the *maximum number of computers*. Here, **SCIENCE block (75 computers)** is the best location.
- **(ii) Hardware Device within Blocks:** Use a **Switch** (preferred) or **Hub** to connect all computers within each block.
- **(iii) Cable Layout:** Draw a minimum spanning tree. Start with the central block (SCIENCE) and connect it to the nearest blocks (LIBRARY at 50m, ARTS at 60m), then connect remaining blocks to the nearest already-connected block.
- **(iv) Router Placement:** Place the router in the block where the server is located (SCIENCE block) or at the main administrative building (ADMIN) for central connectivity.
- **(v) Protocol for Live Lectures:** Use **VoIP (Voice over IP)** or **Video Conferencing protocols** for real-time audio/video communication over the internet.

Practice Question 3: "RetailHub Corp"

"RetailHub Corp" is a retail chain with its headquarters in Bengaluru and a new regional office in Chennai. The Chennai office has four departments: **SALES, LOGISTICS, HR, and MANAGEMENT**. As a network expert, suggest the best network solutions to resolve the issues mentioned in points (i) to (v), keeping the following parameters in mind:

From	To	Distance
SALES	LOGISTICS	30 m
SALES	HR	85 m
SALES	MANAGEMENT	140 m
LOGISTICS	HR	70 m
LOGISTICS	MANAGEMENT	120 m
HR	MANAGEMENT	55 m

Distance of Chennai Office from Bengaluru HQ = 350 km

Block	No. of Computers
SALES	80
LOGISTICS	50
HR	25
MANAGEMENT	10

Questions:

(i) Suggest the most appropriate location of the server inside the Chennai Office. Justify your choice.

(ii) Which hardware device will you suggest to connect all the computers within each block of the Chennai Office?

(iii) Draw a cable layout to efficiently connect various blocks within the Chennai Office.

(iv) Where should the router be placed to provide internet to all the computers in the Chennai Office?

(v) The company wants to set up a secure wired connection between the Chennai Office and the Bengaluru HQ. Which transmission medium (coaxial cable, twisted pair, or optical fiber) will be best suited for this long-distance, high-speed connection? Justify your answer.

> **Help Notes for Practice:**
>
> - **(i) Server Location:** Place the server in the block with the *maximum number of computers*. Here, the **SALES department (80 computers)** is the best location.
> - **(ii) Hardware Device within Blocks:** Use a **Switch** to connect all computers within each block.
> - **(iii) Cable Layout:** Draw a minimum spanning tree. Connect SALES to LOGISTICS (30m), then connect HR to MANAGEMENT (55m), and finally connect the two groups via the next shortest path.
> - **(iv) Router Placement:** Place the router in the block where the server is located (SALES) or at the main entry point for external connectivity.
> - **(v) Transmission Medium:** For a secure, high-speed wired connection over 350 km, use **Optical Fiber**. It offers high bandwidth, is immune to electromagnetic interference, and is ideal for long-distance data transmission.

Practice Question 4: "InnovateTech Labs"

"InnovateTech Labs" is a software development firm with its main branch in Hyderabad and a new development center in Visakhapatnam. The Visakhapatnam center has four blocks: **DEVELOPMENT, TESTING, OPERATIONS, and SERVER**. As a network expert, suggest the best network solutions to resolve the issues mentioned in points (i) to (v), keeping the following parameters in mind:

From	To	Distance
DEVELOPMENT	TESTING	20 m
DEVELOPMENT	OPERATIONS	80 m
DEVELOPMENT	SERVER	110 m
TESTING	OPERATIONS	65 m
TESTING	SERVER	90 m
OPERATIONS	SERVER	40 m

Distance of Visakhapatnam Center from Hyderabad Main Branch = 620 km

Block	No. of Computers
DEVELOPMENT	120
TESTING	40
OPERATIONS	30
SERVER	5

Questions:

(i) Suggest the most appropriate location of the server inside the Visakhapatnam Center. Justify your choice.

(ii) Which hardware device will you suggest to connect all the computers within each block of the Visakhapatnam Center?

(iii) Draw a cable layout to efficiently connect various blocks within the Visakhapatnam Center.

(iv) Where should the router be placed to provide internet to all the computers in the Visakhapatnam Center?

(v) The developers frequently need to send large code files to the Hyderabad branch over the internet. Which protocol (FTP, HTTP, or VoIP) will be used for transferring these files?

Help Notes for Practice:

- **(i) Server Location:** Place the server in the block with the *maximum number of computers*. Here, the **DEVELOPMENT block (120 computers)** is the best location.
- **(ii) Hardware Device within Blocks:** Use a **Switch** to connect all computers within each block.
- **(iii) Cable Layout:** Draw a minimum spanning tree. Connect DEVELOPMENT to TESTING (20m), then connect OPERATIONS to SERVER (40m), and finally connect the two groups via the shortest remaining path (TESTING to OPERATIONS at 65m).
- **(iv) Router Placement:** Place the router in the block where the server is located (DEVELOPMENT) or at the main entry point for external connectivity.
- **(v) Protocol for File Transfer:** Use **FTP (File Transfer Protocol)** for transferring large code files over the internet. (HTTP is for web pages, VoIP is for voice communication).

General Tip for Cable Layout (Part iii):
Draw a graph connecting all the blocks using the minimum total cable length. Find the block with the most connections (central block) and connect it to the nearest blocks, then connect the remaining blocks to the nearest already-connected block (Prim's/Kruskal's algorithm concept).